3.4 Function Notation

Algebra I 3.4 · Student sheet

Name _____ Period ____ Date _____

I can…

- evaluate functions using function notation.
- find the input that gives a certain output.
- graph linear functions and interpret function notation in real life.

Vocabulary · fill in the missing word

Function notation — f(x), read "f of x," is the _____ of function f for the input x

f(a) = b — the point (a, b) is on the _____ of f

Part 1 · Function machine

f(x) = 2x + 5 g(x) = x² − 1

Input x	f(x)	g(x)

Output f(x)	Input x
15	
−3	

Part 2 · Notes: fill in the blanks

f(x) is read "_____."

To evaluate f(3), _____ 3 for x.

To find x when f(x) = 15, set the rule equal to 15 and _____.

f(3) = 11 means the point _____ is on the graph.

Part 3 · Examples (work them with the class)

Example 1

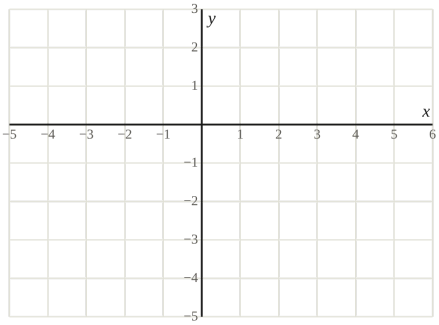
Let f(x) = −4x + 7. Find f(2) and f(−3).

Example 2

Let g(x) = 5x − 3. Find x so that g(x) = 22.

Example 3

Graph f(x) = $\frac{1}{2}$x − 2. Use the graph to find f(4) and the x-value where f(x) = −3.



Example 4

The cost (in dollars) of renting a truck and driving m miles is C(m) = 0.25m + 15. Find C(100) and explain what it means.

Part 4 · Partner practice

Partner 1

Let h(x) = x² + 1. Find h(−3).

Partner 2

Let $f(x) = 3x + 4$. Find x so that $f(x) = -8$.